Math Class 11 Federal Board

Solution of MOST IMPORTANT

Questions of Chapter 7, 8 & 9

Use the method of mathematical induction to prove that

$$P(n): \quad 1.2 + 2.3 + 3.4 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3} \rightarrow (1)$$

Sol: Let the given $P(n)$ be a proposition

Step 1: (Basis step)

put $n=1$ in (1)

$$1(1+1) = \frac{1(1+1)(1+2)}{3}$$

$$1\cdot 2 = \frac{1\cdot 2\cdot \cancel{3}}{\cancel{3}}$$

$$\Rightarrow \quad 2 = 2$$

Thus, $P(1)$ is true.

Step 2: (inductive step)

suppose $P(k)$ is true, so, $P(k+1)$ must be true -

" must be true -

$$P(k): 1\cdot 2 + 2\cdot 3 + 3\cdot 4 + \cdots + k(k+1) = \frac{k(k+1)(k+2)}{3} \rightarrow (2)$$

Add $(k+1)(k+2)$ on both sides of eq. ② we get

$$P(k+1): 1\cdot 2 + 2\cdot 3 + 3\cdot 4 + \cdots + k(k+1) + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$1\cdot 2 + 2\cdot 3 + 3\cdot 4 + \cdots + k(k+1) + (k+1)(\overline{k+1}+1)$$

$$= (k+1)(k+2)\left[\frac{k}{3} + 1\right]$$

$$= (k+1)(k+2)\left[\frac{k+3}{3}\right]$$

$$= \frac{(k+1)(k+2)(k+3)}{3}$$

$$= \frac{(k+1)(\overline{k+1}+1)(\overline{k+1}+2)}{3}$$

Thus $P(k+1)$ is also true.

Thus $P(K+1)$ is also true.
Hence $P(n)$ is true for $n \in Z^+$.

Use the method of mathematical induction to prove that

$$P(n): \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\ldots\left(1-\frac{1}{n+1}\right)=\frac{1}{n+1} \rightarrow (1)$$

Sol: Step-1 put $n=1$,

$$1-\frac{1}{1+1}=\frac{1}{1+1}$$

$$\Rightarrow 1-\frac{1}{2}=\frac{1}{2}$$

$$\Rightarrow \frac{1}{2}=\frac{1}{2}$$

Thus $P(1)$ is true.

Step-2 Assume that $P(K)$ is true
So, $P(K+1)$ must be true

$$(1) \Rightarrow P(K): \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\cdots\left(1-\frac{1}{K+1}\right)=\frac{1}{K+1} \rightarrow (2)$$

multiply by $\left(1-\frac{1}{k+1+1}\right)$ on both sides of Eqn. ②

$P(k+1)$: $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\cdots\left(1-\frac{1}{k+1}\right)\left(1-\frac{1}{k+1+1}\right)$

$$= \frac{1}{k+1}\left(1-\frac{1}{k+1+1}\right)$$

$$= \frac{1}{k+1}\left(1-\frac{1}{k+2}\right)$$

$$= \frac{1}{k+1}\left(\frac{k+2-1}{k+2}\right)$$

$$= \frac{1}{\cancel{(k+1)}}\left(\frac{\cancel{k+1}}{k+2}\right)$$

$$= \frac{1}{k+2} = \frac{1}{k+1+1}$$

Hence $P(k+1)$ is true. Thus, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

$\boxed{10 = 2 \times Q}$ $2\overline{)10}$ 5 ② $\sqrt{10}$ 10 rem

Example 3: Use the method of mathematical induction to show that $n^2 - 3n + 4$ is an even (i.e.; divisible by 2) for all positive integers n.

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Sol $P(n): n^2 - 3n + 4 \rightarrow (1)$

Step-1 put $n = 1$ in (1)

$P(1): 1 - 3 + 4 = 5 - 3 = 2$

Since, 2 is an even No so, $P(1)$ is true.

Step-2 Assume that $P(k)$ is true, so $P(k+1)$ must also be true.

$P(k): k^2 - 3k + 4 = 2Q \rightarrow (2)$
where $Q \in \mathbb{Z}$

For $n = k+1$

$P(k+1): (k+1)^2 - 3(k+1) + 4$

$= k^2 + 1 + 2k - 3k - 3 + 4$

$= (k^2 - 3k + 4) + (2k - 2)$

$= (k^2-3k+4)+(2k-\ldots)$

$= \underline{\underline{2Q}} + \underline{\underline{2(k-1)}}$

$P(k+1)$: $= \underline{\underline{2(Q+k-1)}}$

which is an even no. Thus $P(k+1)$ is also true.

Hence, $P(n)$ is true for all $n \in Z^+$

Example 4: Use the method of mathematical induction to show that $3^n > n^2$ for all positive integers n.

$P(n): 3^n > n^2 \rightarrow (1)$

<u>Sol</u>: <u>Step-1</u> put $n=1$

$3^1 > (1)^2$

$\Rightarrow 3 > 1$

$P(1)$ is true.

<u>Step-2</u> $P(k+1)$ is true when $P(k)$ is true.

Let $p(n)$ is true for $n=k$

$$3^k > k^2 \longrightarrow (2)$$

for $p(k+1)$ we have to show that

for $n=k+1$.

$$3^{k+1} > (k+1)^2 \longrightarrow (3)$$

$$L.H.S = 3^{k+1} = 3^k \cdot 3^1 = 3 \cdot 3^k$$

$$\Rightarrow 3^{k+1} = 3^k + 3^k + 3^k$$

$$3^{k+1} > 3^k + 3^k$$

$3x = 3x$

$x + x + x$

$3^k(1+1+1)$

$3x > 2x$

by (2)

$$\Rightarrow 3^{k+1} > k^2 + 3^k \quad (\because 3^k > k^2)$$

$k=2$) $9 > 4$

Also $3^{k+1} > 2k+1$ for $k > 1$

$k=3$, $27 > 9$

$(k+1)^2$

$k^2 + 2k+1$

$3^{k+1} \Rightarrow k=2$

$$\Rightarrow 3^{k+1} > k^2 + 2k + 1$$

$$\Rightarrow 3^{k+1} > (k+1)^2$$

$\Rightarrow 3^{k+1} > (k+1)^2$ $\quad 3^{k+1} \Rightarrow (k=2)$

$87 > 5$

6427

This shows that $P(k+1)$ is true.

Thus, $P(n)$ is true for all $n \in Z^+$.

Show that the inequality $4^n > 3^n + 4$ is true, for integral values of $n \geq 2$.

Sol Step-1 put $n=2$ $\quad P(n): 4^n > 3^n + 4$

$$4^2 > 3^2 + 4$$

$$16 > 13$$

$P(1)$ is true.

Step suppose $P(k)$ is true

put $n=k \Rightarrow \quad 4^k > 3^k + 4 \rightarrow (2)$

So, $P(k+1)$ must be true if $P(k)$ is true.

So, P(k) is true.

for $n=k+1$, we have to showed

$$4^{k+1} > 3^{k+1}+4 \rightarrow (3)$$

known $\quad 4^k > 3^k+4$

$\times$ by 4 on both sides

$$4 \cdot 4^k > 4(3^k+4)$$

$$\Rightarrow 4^{k+1} > 4 \cdot 3^k + 16$$

$$\Rightarrow 4^{k+1} > (3+1) \cdot 3^k + 16$$

$$\Rightarrow 4^{k+1} > 3 \cdot 3^k + 3^k + 16$$

$9 > 5+3$

$$\Rightarrow 4^{k+1} > 3^{k+1} + \underline{3^k + 4 + 12}$$

$$\Rightarrow 4^{k+1} > 3^{k+1} + 4 + \underline{(3^k+12)}$$

$\therefore 3^k+12 > 0$

$$\Rightarrow 4^{k+1} > 3^{k+1} + 4$$

So, $P(k+1)$ is true, Hence $P(n)$ is true for $n \geq 2$.

Example 5: Use the method of mathematical induction to show that non-negative

$4 + 4.6 + 4.6^2 + 4.6^3 + \cdots + 4.6^n = \frac{4(6^{n+1}-1)}{5}$ for all positive integers n.

Step-1 put $n = 0$,

$$4 \cdot 6^0 = \frac{4(6^1 - 1)}{5}$$

$$4 = \frac{4 \cdot \cancel{5}}{\cancel{5}}$$

$$\Rightarrow 4 = 4$$

So, $P(1)$ is true.

Step-2 $P(k)$ is true say

$n = k \Rightarrow$

$$4 + 4 \cdot 6 + \cdots + 4 6^k = \frac{4(6^{k+1} - 1)}{5} \rightarrow (1)$$

So, $P(k+1)$ must be true

$n = k+1$ Adding $4\cdot 6^{k+1}$ term on both sides of ①, we get

$$4 + 4\cdot 6 + 4\cdot 6^2 + \cdots + 4\cdot 6^k + 4\cdot 6^{k+1} = \frac{4(6^{k+1}-1)}{5} + 4\cdot 6^{k+1}$$

$$= \frac{4(6^{k+1}-1) + 5\cdot 4\cdot 6^{k+1}}{5}$$

$x + 5x$
$= 6x$

$$= \frac{4\left[6^{k+1} - 1 + 5\cdot 6^{k+1}\right]}{5}$$

$$= \frac{4(6^{k+1} + 5\cdot 6^{k+1} - 1)}{5}$$

$$= \frac{4((1+5)6^{k+1} - 1)}{5}$$

$$= \frac{4(6^1\cdot 6^{k+1} - 1)}{}$$

$$= \frac{4(6^{k+1+1}-1)}{5}$$

This shows that $P(K+1)$ is true.

Hence, $P(n)$ is true for all $n \in Z^{+} \cup \{0\}$ ↳ Non-negative integer

$\forall\, n \geq 0$

xii.	If $A = \begin{bmatrix} x & 0 \\ y & 1 \end{bmatrix}$, then show that $A^n = \begin{bmatrix} x^n & 0 \\ \frac{y(x^n-1)}{(x-1)} & 1 \end{bmatrix}$, $n \epsilon z^+$

Apply the principle of Mathematical Induction to prove that $7^{2n} + 7$ is divisible by 8 for all positive integral values of n.

$P(n): 7^{2n}+7$ ①

Sol Step-1 put $n=1$

$$7^{2(1)}+7 = 49+7=56$$

... divisible by 8 So,

As 56 is divisible by 8 so,

P(1) is true.

Step-2 Assume P(K) is true

so, P(K+1) must also be true.

P(K) is divisible by 8.

$P(K) = 7^{2K}+7 = 8Q$ (2)

$\Rightarrow 7^{2K} = 8Q-7$

For $n=K+1$

$$7^{2(K+1)}+7 = 7^{2K+2}+7$$

$$= 7^{2K}\cdot 7^2+7$$

$$= (8Q-7)49+7$$

$$= 8\cdot 49Q - 343 + 7$$

$$= 8\cdot 49Q - 336$$

49

7

343

$= 8 \cdot 449 - 336$

$= 8 \cdot 449 - 8 \cdot 42$

$= 8\underline{(449 - 42)}$

which is clearly divisible by 8.

So, $P(k+1)$ is true. Hence

$P(n)$ is ______ $n \in Z^+$

Find the constant term in the expansion of $\left(x + \frac{2}{x}\right)^{10}$.

$\binom{n}{r}_{n-(n-1)}$

$(a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \ldots$

$+ \binom{n}{n-1} a^1 b^{n-1}$

$+ \binom{n}{n} a^0 b^n$

$n \geq 0 \rightarrow \underline{n \in Z^+}$

Total terms in $(a+b)^n$ expansion $= n+1$

$T_1 \rightarrow 1^{st}$ term

General Term

$\binom{n}{} \quad {}^{n-r}\ {}^{r}$

$$\boxed{T_{r+1} = \binom{n}{r} a^{n-r} b^{r}}$$

$5 = 5 \cdot x^0$

10, 9, 8, 7, 6, 5

0, 1, 2, 3, 4, 5

Find the constant term in the expansion of $\left(x + \frac{2}{x}\right)^{10}$.

$10 \to n$

$2(x^{-1})$

$\boxed{5 \quad -5}$

Term independent of x?

$\boxed{x^0}$

Sol

$n = 10, \quad a = x, \quad b = \frac{2}{x}$

$$T_{r+1} = \binom{n}{r} a^{n-r} b^{r}$$

$$T_{r+1} = \binom{10}{r} x^{10-r} \cdot \left(\frac{2}{x}\right)^{r}$$

$$T_{r+1} = \binom{10}{r} \frac{2^r}{x^r} \cdot x^{10-r}$$

$$T_{r+1} = \binom{10}{r} 2^r \cdot x^{-r} \cdot x^{10-r}$$

$$T_{r+1} = \binom{10}{r} 2^r \cdot x^{10-2r} \to 1$$

$n-r-r$

$-2-1=-3$

$2+1=3$

$1-2=-1$

For constant term $\quad 10 - 2r = 0$

For constant term $10 - 2r = 0$

$\Rightarrow 2r = 10$

$\boxed{r = 5}$

$T_{5+1} = \binom{10}{5} 2^5 x^{10-10}$

$\boxed{T_6 = 8064}$ Ans

Example 10: Find the 3rd term from the end in the expansion of $\left(2 - \frac{5}{\sqrt{x}}\right)^5$.

$n = 5$

$r = 3^{rd}$ term from End

1, 2, 3, 4, 5, 6

$r = 3$

Term # from beginning

$= (n - r + 2)^{th}$ term

$= (5 - 3 + 2)^{th}$ term

$= 4^{th}$

$r + 1 = 4$

T_4 ? $n = 5$, $r = 3$, $a = 2$, $b = \frac{-5}{\sqrt{x}}$

$T_{r+1} = \binom{n}{r} a^{n-r} \cdot b^r$

$T_{r+1} = \left(\;\right)$

$$T_4 = \binom{5}{3} 2^2 \left(-\frac{5}{\sqrt{x}}\right)^3$$

$$T_4 = \underline{\underline{\binom{5}{3} \times 4 \times (-5)^3}} \cdot \left(\frac{1}{\sqrt{x}}\right)^3$$

$$T_4 = -5000 \frac{1}{(x^{1/2})^3}$$

$$= -\frac{5000}{x^{3/2}} = -5000\, x^{-\frac{3}{2}}$$

$$\boxed{T_4 = -5000\, x^{-\frac{3}{2}}} \quad \text{As}$$

$\rightarrow$ 3rd Term from end

Example: The fourth term in the expansion of $\left(ax + \frac{1}{x}\right)^n$ is $\frac{5}{2}$

Find the values of a and n.

Sol

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r \rightarrow (1)$$

Given $r+1 = 4 \Rightarrow \boxed{r = 3}$

$\Rightarrow T_4 = \frac{5}{2}$, $a' = ax$, $b = \frac{1}{x}$

$$\frac{5}{2} = \binom{n}{3}(ax)^{n-3} \cdot \left(\frac{1}{x}\right)^3$$

$$\frac{5}{2} = \binom{n}{3} a^{n-3} \cdot x^{n-3} \cdot \frac{1}{x^3}$$

$$\frac{5}{2} = \binom{n}{3} a^{n-3} \cdot x^{n-6} \rightarrow (2)$$

As T_4 is a constant term so,

$$x^{n-6} = x^0$$

$$n-6 = 0 \Rightarrow \boxed{n=6}$$

$$(2) \Rightarrow \quad \frac{5}{2} = \binom{6}{3} a^{6-3} \Rightarrow \frac{5}{2} = 20a^3$$

$$a^3 = \frac{\frac{5}{2}}{20} = \frac{1}{8} =$$

$$(\ ?\ x)\ \frac{1}{x} = \left(\frac{1}{8}\right)^{1/3}$$

$$\Rightarrow (a^3)^{\frac{1}{3}} = \left(\frac{1}{8}\right)^{\frac{1}{3}}$$

$$\boxed{a = \frac{1}{2}}$$

Middle Term $(a+b)^n$

Case 1 n is Even ($n=4$)

Total terms = odd

1 2 3

3, 7, 5, 11, 13

2, 3, 4, 5, 10

Middle Term = $\left(\frac{n}{2}+1\right)^{th}$

$n=4 \Rightarrow \left(\frac{4}{2}+1\right)^{th} = 3^{rd}$ te

When n is even $\rightarrow$ $\left(\frac{n+2}{2}\right)^{th}$ term

$$\left(3 - \frac{x}{4}\right)^8$$

Middle Term

$(8+1)^{th}$ te

Middle term = $\left(\frac{8}{2}+1\right)^{th}$ term

= 5^{th} term

T_5 = middle

$T_{r+1} = \binom{n}{r} a^{n-r} b^r$

$T_{4+1} = \binom{8}{4} 3^4 \cdot \left(-\frac{x}{4}\right)^4$ ✓

Case 2 n is odd (e.g. $n = 7$)

↓ total terms = Even

$\left(x - \frac{1}{x}\right)^7$

There are 2 middle terms when n is odd

1, 3, 5, 7

T_2 T_3

1, 2, 3

Even

$\left(\frac{n+1}{2}\right)^{th}$ and $\left(\frac{n+3}{2}\right)^{th}$ term

Middle terms $\left(\frac{7+1}{2}\right)$, $\left(\frac{7+3}{2}\right)^{th}$

4^{th}, 5^{th} term

T_4 , T_5

$r=3$, $r=4$

~~1, 2, 3~~ 4, 5 $\boxed{6, 7, 8}$

7.3

Binomial Series

$(x+\frac{1}{x})^5$ $\frac{1}{(x+\frac{1}{x})^{-5}}$

$(a+b)^{-ve \text{ or } fraction} \Rightarrow$ infinite series

convergent (finite)

$S_\infty = \frac{a}{1-r}$

Divergent (infinite)

$S_n = \frac{a(1-r^n)}{1-r}$

~~10~~ < 100

$|r| < 1$

$\pm r < 1$

$1 < 2$ $-1 > -2$

$\boxed{r < 1}$

$-r < 1$

$\boxed{r > -1}$ | $\boxed{-1 < r < 1}$

$$\boxed{r > -1} \quad \Big| \quad \boxed{-1 < r < 1}$$

$K \leq 2$ $(1 > 2)$ Rec.

$|r| > 1 \qquad \pm r > 1$

$-r > 1 \qquad r > 1$

$r < -1 \qquad r > 1$

$\binom{n}{r-1}$

negative

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$\boxed{T_{r+1} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}x^r}$$

(iii). $\left(\frac{5}{2} - \frac{3}{x^2}\right)^{\frac{1}{2}}$

$$= \left(\frac{5}{2}\left(\frac{\frac{5}{2} - \frac{3}{x^2}}{\frac{5}{2}}\right)\right)^{\frac{1}{2}}$$

$$\boxed{\begin{aligned} &\left|-\frac{6}{5x^2}\right| < 1 \\ &\left|\frac{1}{x^2}\right| < \frac{5}{6} \\ &\boxed{|x^2| > \frac{6}{5}} \end{aligned}}$$

$$= \left(\frac{5}{2}\left(1-\frac{\frac{3}{x^2}}{\frac{5}{2}}\right)\right)^{\frac{1}{2}}$$

$\frac{3}{x^2} \div \frac{5}{2}$

$\frac{3}{x^2} \times \frac{2}{5}$

$$= \left(\frac{5}{2}\right)^{\frac{1}{2}}\left(1-\frac{6}{5x^2}\right)^{\frac{1}{2}}$$

$$= \left(\frac{5}{2}\right)^{\frac{1}{2}}\left(1+\left(\frac{-6}{5x^2}\right)\right)^{\frac{1}{2}}$$

$$= \left(\frac{5}{2}\right)^{\frac{1}{2}}\left[1-\frac{1}{2}\cdot\frac{6}{5x^2}+\frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2}\left(\frac{-6}{5x^2}\right)^2+\dots\right]$$

$$\left|\frac{-6}{5x^2}\right|<1 \Rightarrow \frac{6}{5x^2}<1$$

$$\frac{1}{x^2}<\frac{5}{6}$$

$$\boxed{x^2>\frac{6}{5}}$$

If x is very small such that its square and higher powers can be neglected, then show that :

$$\frac{(16+4x)^{\frac{3}{4}}}{(4+x)\sqrt{9-6x}} \approx \frac{2}{3}+\frac{13x}{72}$$

2025 1st

7.4

$$L.H.S = \frac{(16+4x)^{\frac{3}{4}}}{(4+x)(\sqrt{9-6x})}$$

$$= \frac{\left(16\left(1+\frac{4x}{16}\right)\right)^{\frac{3}{4}}}{4\left(1+\frac{x}{4}\right)\left(9\left(1-\frac{6x}{9}\right)\right)^{\frac{1}{2}}}$$

$$= \frac{16^{\frac{3}{4}}\left(1+\frac{x}{4}\right)^{3/4}}{4\left(1+\frac{x}{4}\right)9^{1/2}\cdot\left(1-\frac{6x}{9}\right)^{\frac{1}{2}}}$$

$$= \frac{8\left(1+\frac{3}{4}\cdot\frac{x}{4}+\cdots \text{Neglected}\right)}{3\times\cancel{4}\left(1+\frac{x}{4}\right)\left(1-\frac{1}{\cancel{2}}\cdot\frac{\overset{3}{\cancel{6}}x}{\underset{3}{\cancel{9}}}\right)}$$

$$2\left(1+\frac{3x}{16}\right)$$

$$= \frac{2\left(1+\frac{3x}{16}\right)}{3\left(1+\frac{x}{4}\right)\left(1-\frac{x}{3}\right)}$$

$$= \frac{2}{3}\,\frac{\left(1+\frac{3x}{16}\right)}{1-\frac{x}{3}+\frac{x}{4}+\text{neglect}}$$

$$= \frac{2}{3}\cdot\frac{\left(1+\frac{3x}{16}\right)}{\left(1-\frac{x}{12}\right)}$$

$$= \frac{2}{3}\left(1+\frac{3x}{16}\right)\left(1-\frac{x}{12}\right)^{-1}$$

$$= \frac{2}{3}\left(1+\frac{3x}{16}\right)\left(1+\frac{x}{12}\right)$$

$$= \frac{2}{3}\left[1 + \frac{3x}{16} + \frac{x}{12}\right]$$

$$= \frac{2}{3}\left[1 + \frac{13x}{48}\right]$$

$$= \frac{2}{3} + \frac{\cancel{2}}{3} \cdot \frac{13x}{\cancel{48}\ 24}$$

$$= \frac{2}{3} + \frac{13x}{72} = RHS$$

Find the unit digit of: (i) 17^{203} (ii) 29^{26} (iii) 36^{307}

i) 17^{203} The unit digit depends on the unit digit of the base

$17^{203} \approx 7^{203}$

As $7^1 = 7$, $7^2 = 49$, $7^3 = 343$, $7^4 = 2401$

The cycle of 7^n is $\boxed{7, 9, 3, 1}$

203

The cycle of 7 ...

The cycle length of 7^{203} is $\underline{\underline{4}}$

$$4\overline{)203}$$
quotient 50
$$-200$$
$$3$$

$$203 = 4 \times 50 + \underline{\underline{3}}$$

$$\Rightarrow 7^{203} \sim 7^3 = 34\underline{\underline{3}}$$

$\boxed{\text{Then the unit place of } 17^{203} = 3}$

(ii) $29^{26} = 9^{26}$

$9^1 = 9, \quad 9^2 = 8\underline{\underline{1}}$

cycle length of 9 is 2

$$2\overline{)26}$$
quotient 13
$$26$$
$$0$$

Thus the unit digit of $29^{26} = 1$

... 20 ... is even number

Or since 28 is even number

So, unit digit of $29^{26} = 1$

(c) $36^{307} = 6^{307}$

$6^1 = 6$, $\boxed{\text{cycle length} = 1}$

Thus the unit digit of $6^{307} = 6$

Check Point

Find the unit digit of $8^{48} + 6$??

So cycle length of 8^{48}

$8^1 = 8$, $8^2 = 64$, $8^3 = 512$, $8^4 = 4096$

$\boxed{8, 4, 2, 6}$

cycle length of $8 = 4$

cycle length of $8 = 4$

$$4 \overline{)48}^{12} \quad \frac{48}{0} \to R$$

Thus 8^{48} has unit digit is 6

$$8^{48} + 6 = 6 + 6 = 12$$

$$10 \overline{)12}^{1} \quad \frac{10}{2}$$

$\boxed{\text{Thus, the unit digit of } 8^{48} + 6 = 2}$

Find the last two digits of the number $(13)^{10}$.

Sol $10 = 2 \times 5$

$$(13)^{10} = \left((13)^2\right)^5$$

$$= (169)^5$$

$$= (170 - 1)^5$$

$$10 \overline{)543}^{54} \quad \frac{50}{43} \quad \frac{40}{3}$$

$$100 \overline{)543}^{5} \quad \frac{500}{43}$$

$$(170-1)^5 = \binom{5}{0} 170^5 + \binom{5}{1} 170^4 (-1) + \cdots$$

$$(170-1)^5 = \binom{5}{0}170^5 + \binom{5}{1}170^4(-1) + \cdots + \binom{5}{4}170^1(-1)^4 + \binom{5}{5}(-1)^5$$

$10\overline{)26}$, 20, 6 ; $= 10 \cdot 2 + 6$

$$(13)^{10} = (170-1)^5 = \underline{100K} + \binom{5}{4}170 - 1 \quad , K \in \mathbb{Z}$$

$$(13)^{10} = \underline{849} \quad 100\overline{)849}, \ 800, \ 49$$

$\boxed{\text{Thus, last two digits of } 13^{10} \text{ are } 49}$

Q Find the last 2 digits of

$$(23)^{14} ?$$

$$= \left((23)^2\right)^7 = (529)^7$$

$$(23)^{14} = (500+29)^7$$

$(29)^7$

$$= \binom{7}{0}500^7 + \binom{7}{1}500^6\,29 + \cdots + \binom{7}{6}500^1\,29^6 + \binom{7}{7}29^7$$

$$(23)^{14} = 100k + \underline{\underline{29^7}} \quad k \in \mathbb{Z}$$

Last two digits of $(23)^{14}$ depend on 29^7

$29^1 = \underline{\underline{29}}$

$29^2 = 29 \times 29 = 8\boxed{41}$

$29^3 = 29 \times 41 = 11\,89$

$29^4 = 29 \times 89 = 25\,81$

$29^5 = 29 \times 81 = 23\boxed{49}$

$29^6 = 29 \times 49 = 14\boxed{21}$

$29^7 = 29 \times 21 = 6\boxed{09}$

$$\begin{array}{r} 8 \\ 100\,\overline{)\,841} \\ 800 \\ \hline 41 \end{array}$$

$$\begin{array}{r} 6 \\ 100\,\overline{)\,609} \\ 600 \\ \hline \boxed{09} \end{array}$$

$$\boxed{\text{Thus last two digits of } (23)^{14} = 09}$$

By Binomial Theorem

Find the remainder when 7^{101} is divided by 25.

$$4\overline{)24}\ 6,\quad 24,\quad \alpha$$

$$\frac{25}{4} = \frac{\overbrace{4\times 6}^{Q} + \overbrace{1}^{R}}{4} \qquad 4\overline{)25}\ 6,\quad 24,\quad 1 \text{ Remainder}$$

$$= \underset{\text{Quo}}{6} + \frac{1}{4} \to \text{Div}$$

$LCM = 2\times 50$

Sol

$$\frac{7^{101}}{25} = \frac{7\cdot 7^{100}}{25} = \frac{7(7^2)^{50}}{25}$$

$$\frac{7^{101}}{25} = \frac{7(49)^{50}}{25} = \frac{7(50-1)^{50}}{25}$$

$$\frac{7^{101}}{25} = \frac{7(50+(-1))^{50}}{25} \quad \text{By using Binomial Theorem}$$

$$7^{101} = 7\left[\binom{50}{0}50^{50} + \binom{50}{1}50^{49}(-1) + \dots + \binom{50}{49}50^{1}(-1)^{49} + \binom{50}{50}(-1)^{50}\right]$$

$$\frac{7}{25} = \frac{7}{25}\left[\binom{50}{0}50 + \binom{50}{1}50(-1) + \cdots + (50)^{(-1)}\right]$$

$$\frac{7^{101}}{25} = \frac{7}{25}\left[25k+1\right]$$

$$= \frac{7\times 25k}{25} + \frac{7}{25}$$

$$= \frac{25\times 7k}{25} + \frac{7}{25}$$

Quot

$$\frac{7^{101}}{25} = \frac{25\times\boxed{7k}+7}{25}, \quad k\in\mathbb{Z}$$

Thus, the Remainder of $\frac{7^{101}}{25}$ is

7 Az

Find the remainder when 7^{103} is divided by 25.

$$\frac{7^{103}}{25} = \frac{7\cdot 7^{102}}{25} = \frac{7\cdot(7^2)^{51}}{25} = \frac{7(50-1)^{51}}{25}$$

$$\frac{7^{103}}{25} = \frac{7}{25}\left[\binom{51}{0}50^{51} + \cdots + \binom{51}{50}50^{1}(-1)^{50} + \binom{51}{51}(-1)^{51}\right]$$

$$= \frac{7}{25}\left[250K - 1\right] \quad K \in \mathbb{Z}$$

$$= \frac{175K - 7}{25} = \frac{175K - 25 + 25 - 7}{25} = \frac{25(7K-1) + 18}{25}$$

1st method Remainder $= -7 + 25 = 18$

2nd method $\cancel{\frac{25K + 7 - 7}{25}}$

$\cancel{= 25K + 25 + 25}$

If the fractional part of the number $\frac{2^{403}}{15}$ is $\frac{k}{15}$, then find k.

Sol

$403 \div 4 = 100$, $4 \times 100 = 400$, remainder 3

$\Rightarrow 403 = 4 \times 100 + 3$

$2^{403} \quad 2^{400} \cdot 2^{3} = (2^{4})^{100} \cdot 8$

$$\frac{2^{403}}{15} = \frac{2^{400}\cdot 2^{3}}{15} = \frac{(2^{4})^{100}\cdot 8}{15}$$

$$\frac{2^{403}}{15} = \frac{8}{15}(16)^{100} = \frac{8}{15}(15+1)^{100}$$

$$\frac{8}{15}(15+1)^{100} = \frac{8}{15}\left[\binom{100}{0}15^{100} + \binom{100}{1}15^{99} + \cdots + \binom{100}{99}15^{1} + \binom{100}{100}\right]$$

$$= \frac{8}{15}(15\lambda + 1) \quad \lambda \in Z^{+}$$

$$= \frac{8\times 15\lambda}{15} + \frac{8}{15}$$

$$\frac{2^{403}}{15} = \frac{15\times 8\lambda}{15} + \left\{\frac{8}{15}\right\}$$

Thus the fractional part of

$$\frac{2^{403}}{15} = \frac{8}{15} = \frac{k}{15}$$

$\left(8 + \frac{27}{3}\right)^{133}$ $\quad \boxed{k=8}$

$8 + 27 \; 3^3$

$2 \; 1^3$

5^3

$\Rightarrow \boxed{k=8}$

Which one is greater $99^{50} + 100^{50}$ or 101^{50}?

$99 \quad 100 \quad 101$

Sol

$(101)^{50} = (100+1)^{50}$

$= \binom{50}{0}100^{50} + \binom{50}{1}100^{49} + \binom{50}{2}100^{48} + \dots \to (1)$

$(99)^{50} = (100-1)^{50} = \binom{50}{0}100^{50} - \binom{50}{1}100^{49} + \binom{50}{2}100^{48} + \dots \to (2)$

Subtract (2) from (1), we

$(101)^{50} - (99)^{50} = \underline{50 \times 100^{49}} + \underline{50 \times 100^{49}} + \dots$

$= 2 \cdot 50 \times 100^{49}$

$(101)^{50} - (99)^{50} = 100^1 \times 100^{49} +$ more than double

$(101)^{50} - (99)^{50} = 100^{50} +$ more than double

$(101)^3 - (99)^3 = 100$ If more the dust

$9 - 5 = 2 + 1$

$4 \quad 7 \quad 3$

472

$9 - 12 = -3$

$(101)^{50} - 99^{50} > 100^{50}$

$(101)^{50} > (99)^{50} + 100^{50}$

Thus 101^{50} is greater than

$99^{50} + 100^{50}$

$y = 2x + 3$

$y > x$

$x > y$

$y > 2x$

$y = \boxed{0.3}x$

$\frac{3}{10}x$

$x = 1$

$y = 5$

$x = 1$

$y = 0.3$

Show that $11^9 + 9^{11}$ is divisible by 10.

$$11^9 + 9^{11} = (10+1)^9 + (10-1)^{11}$$

$$= \binom{9}{0}10^9 + \binom{9}{1}10^8 + \cdots + \binom{9}{8}10 + \binom{9}{9} + \binom{11}{0}10^{11} - \binom{11}{1}10^{10} + \cdots + \binom{11}{10}10 + \binom{11}{11}(-1)$$

$$= 10^9 + 9\cdot 10^8 + \cdots + 9\cdot 10 + \cancel{1} + 10^{11} - 11\cdot 10^{10} + \cdots + 11\cdot 10 \cancel{-1}$$

$$= 10k, \quad k \in \mathbb{Z}$$

As $11^9 + 9^{11}$ is a multiple of 10 Thus, this no. is divisible by 10 Ans

Check Point

$\frac{5^{99}}{13} \rightarrow$ Find its Remainder

$$\frac{5^{99}}{13} = \frac{5\cdot 5^{98}}{13} = \frac{5\cdot (5^2)^{49}}{13}$$

Sol $\frac{5}{13} = \frac{\ldots}{13} \ldots \frac{\ldots}{13}$

$$= \frac{5}{13}\left(26-1\right)^{49}$$

$$= \frac{5}{13}\left[\binom{49}{0}26^{49} + \binom{49}{1}26^{48}(-1) + \cdots + \binom{49}{48}26 + \binom{49}{49}(-1)^{49}\right]$$

$$= \frac{5}{13}\left[13K - 1\right] \quad K \in Z.$$

$$= \frac{5 \times 13K - 5}{13}$$

$$\boxed{\begin{aligned}-5+13\\ =8\end{aligned}}$$

$$= \frac{13 \times 5K - 13 + 13 - 5}{13}$$

$$= \frac{13(5K-1) + 8}{8}$$

Thus $\frac{5^{99}}{13}$ has a Remainder of $\boxed{8}$ Ans